

NAMSOS, Norway

WMO: 012900

Lat: 64.471N

Lon: 11.571E

Elev: 2

StdP: 101.30

Time Zone: 1.00 (EUC)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.2	-11.9	-17.3	0.8	-12.9	-15.3	1.0	-10.3	14.8	2.3	13.4	-0.1	6.1	50	0.698

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	26.0	17.6	23.7	16.5	21.4	15.4	18.7	24.7	17.2	22.5	15.9	20.2	3.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	11.4	21.5	14.9	10.6	19.6	13.8	9.8	18.2	52.9	25.0	48.3	22.4	44.6	20.2	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 11.5	10.1	9.0		-17.2	29.3	2.8	2.7	-19.2	31.2	-20.9	32.8	-22.5	34.4	-24.5	36.4	
(5)			DB	-17.5	20.1	2.8	1.9	-19.5	21.4	-21.1	22.5	-22.7	23.6	-24.8	24.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-2.2	-1.2	0.4	4.0	8.8	11.7	15.1	14.2	10.9	5.7	1.8	-0.1
	DBStd	7.07	4.64	4.66	4.22	3.25	4.14	3.08	3.99	2.81	3.02	3.01	3.74	4.47
	HDD10.0	1988	377	315	299	182	74	16	1	1	24	138	247	314
	HDD18.3	4601	635	548	557	428	296	200	118	131	223	393	497	572
	CDD10.0	453	0	0	0	3	37	66	160	132	51	4	0	0
	CDD18.3	24	0	0	0	0	1	1	19	4	0	0	0	0
	CDH23.3	233	0	0	0	0	14	4	181	33	1	0	0	0
	CDH26.7	55	0	0	0	0	1	0	53	1	0	0	0	0
(14) Wind	WSAvg	4.4	5.8	5.4	4.8	4.1	3.7	3.5	3.2	3.3	4.2	4.6	5.0	5.2
(15) Precipitation	PrecAvg	1294	137	113	119	74	64	84	87	86	129	133	120	138
	PrecMax	1670	281	303	280	154	105	162	187	143	313	228	252	288
	PrecMin	865	4	19	30	20	26	9	30	21	32	37	15	55
	PrecStd	175	76	65	63	34	20	37	38	37	74	46	72	61
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	7.7	10.0	18.5	25.3	23.9	31.1	26.5	21.2	16.7	10.3	8.8
		MCWB	5.8	5.3	5.4	10.2	15.2	14.8	20.4	17.5	15.0	11.7	8.7	5.8
	2%	DB	6.0	6.8	7.8	14.4	22.3	21.3	28.0	24.3	19.0	13.1	8.9	7.5
		MCWB	4.5	4.5	4.3	7.8	14.1	14.4	18.6	16.7	14.1	9.8	7.6	5.2
	5%	DB	4.8	5.9	6.5	11.5	19.3	19.4	24.9	21.8	17.1	11.1	7.6	6.5
		MCWB	3.2	4.0	4.2	6.0	12.6	13.7	18.2	15.8	12.9	8.6	6.3	4.7
	10%	DB	3.7	4.7	5.6	9.2	16.2	17.6	22.6	19.7	15.6	9.7	6.2	5.3
		MCWB	2.2	3.0	3.7	5.2	11.3	12.9	16.8	14.9	12.0	7.8	4.8	3.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	6.1	6.6	10.8	16.2	16.9	21.5	18.7	15.7	12.4	9.2	7.3
		MCDB	7.1	7.4	8.0	18.1	23.3	22.4	28.8	24.4	19.8	15.6	9.7	7.8
	2%	WB	4.9	5.2	5.5	8.5	14.5	15.4	19.8	17.3	14.4	10.2	7.8	6.0
		MCDB	5.5	6.3	6.8	14.0	20.9	19.8	26.2	22.6	18.0	12.1	8.7	6.7
	5%	WB	3.3	4.3	4.7	6.7	13.0	14.1	18.5	16.2	13.4	9.0	6.3	4.8
		MCDB	4.4	5.6	6.3	10.1	18.8	18.1	24.4	20.6	16.4	10.8	7.2	6.0
	10%	WB	2.3	3.0	3.7	5.6	11.5	13.0	17.0	15.2	12.5	8.1	5.1	3.6
		MCDB	3.3	4.6	5.4	8.5	16.1	17.1	22.0	19.0	15.0	9.4	6.0	5.1
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	4.6	6.1	7.5	8.3	6.9	7.9	7.7	6.4	5.2	4.1	4.8
		MCDBR	5.1	5.7	6.3	11.8	13.8	11.2	13.3	11.7	9.4	7.2	5.2	5.8
		MCWBR	4.2	4.3	4.4	6.6	7.0	5.7	6.1	5.7	5.2	4.6	4.2	4.7
	5% WB	MCDBR	4.9	5.4	5.7	9.8	13.3	10.2	12.5	10.6	8.6	6.0	5.1	5.7
		MCWBR	4.2	4.6	4.4	5.9	7.1	5.4	6.0	5.6	5.0	4.2	4.5	5.0
(40) Clear-Sky Solar Irradiance	taub		0.381	0.301	0.330	0.357	0.356	0.355	0.362	0.349	0.335	0.311	0.345	0.364
	taud		2.600	2.178	2.239	2.334	2.422	2.459	2.466	2.493	2.516	2.388	2.513	2.507
	Ebn at Noon		293	589	743	815	854	862	844	822	755	593	286	116
	Edh at Noon		26	59	85	98	98	97	94	83	67	48	23	14
(44) All-Sky Solar Radiation	RadAvg		0.11	0.55	1.72	3.41	4.57	4.88	4.61	3.54	2.02	0.87	0.20	0.04
	RadStd		0.01	0.09	0.30	0.32	0.30	0.61	0.48	0.37	0.24	0.13	0.04	0.00

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	-143	-189	+67	N/A

Nomenclature: See separate page